Table of Z-Transforms

	f[n]	F(z).	Region of Convergence
1.	$\delta[n]$	1	
2	$\delta[n-m]$	z^{-m}	
3	$a^n u_0[n]$	$\dfrac{z}{z-a}$	$\mid z \mid > a$
4	$u_0[n]$	$\dfrac{z}{z-1}$	$\mid z \mid > 1$
5	$(e^{-anT_s})u_0[n]$	$\dfrac{z}{z-e^{-aT_s}}$	$\mid e^{-aT_s}z^{-1} \mid < 1$
6	$(\cos naT_s)u_0[n]$	$\dfrac{z^2 - z\cos aT_s}{z^2 - 2z\cos aT_s + 1}$	$\mid z \mid > 1$
7	$(\sin naT_s)u_0[n]$	$\dfrac{z\sin aT_s}{z^2 - 2z\cos aT_s + 1}$	$\mid z \mid > 1$
8	$(a^n \cos naT_s)u_0[n]$	$\dfrac{z^2 - az\cos aT_s}{z^2 - 2az\cos aT_s + a^2}$	$\mid z \mid > 1$
9	$(a^n \sin naT_s)u_0[n]$	$\dfrac{az\sin aT_s}{z^2 - 2az\cos aT_s + a^2}$	$\mid z \mid > 1$
10	$u_0[n] - u_0[n-m]$	$\dfrac{z^m - 1}{z^{m-1}(z-1)}$	
11	$nu_0[n]$	$\dfrac{z}{(z-1)^2}$	
12	$n^2 u_0[n]$	$\dfrac{z(z+1)}{(z-1)^3}$	
13	$[n+1]u_0[n]$	$\dfrac{z^2}{(z-1)^2}$	
14	$a^n n u_0[n]$	$\dfrac{az}{(z-a)^2}$	
15	$a^n n^2 u_0[n]$	$\dfrac{az(z+a)}{(z-a)^3}$	
16	$a^n n[n+1]u_0[n]$	$\dfrac{2az^2}{(z-a)^3}$	

See also: Wikibooks: Engineering_Tables/Z_Transform_Table and Z-Transform—WolframMathworld for more complete references.